

Anesthesia for Central Nervous System Diseases – MCQ

Source: Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Central Nervous System Diseases

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. In a patient with Parkinson's disease, which drug should be continued on the morning of surgery?

- A. Anticholinergics
- B. Levodopa/Carbidopa
- C. Benzodiazepines
- D. MAO-B inhibitors

Ans: B

Q2. Which anesthetic agent should be AVOIDED in Parkinson's disease due to its antidopaminergic effects?

- A. Propofol
- B. Ketamine
- C. Droperidol
- D. Thiopentone

Ans: C

Q3. Parkinson's disease patients are at increased risk of which respiratory complication?

- A. Bronchospasm
- B. Aspiration pneumonia
- C. Pneumothorax
- D. Pulmonary embolism

Ans: B

Q4. Which interaction is most dangerous in a patient taking Selegiline (MAO-B inhibitor) for Parkinson's disease?

- A. Propofol causing bradycardia
- B. Meperidine (Pethidine) causing serotonin syndrome
- C. Succinylcholine causing hyperkalemia
- D. Atracurium causing histamine release

Ans: B

Q5. In Alzheimer's disease, which enzyme inhibitor is used for treatment and may prolong the action of succinylcholine?

- A. MAO inhibitors
- B. Anticholinesterase inhibitors (Donepezil)
- C. ACE inhibitors
- D. Proton pump inhibitors

Ans: B

Q6. Which of the following is a primary anesthetic concern in Alzheimer's disease?

- A. Malignant hyperthermia
- B. Postoperative cognitive dysfunction (POCD)
- C. Autonomic hyperreflexia
- D. Latex allergy

Ans: B

Q7. Which anticholinergic is preferred in Alzheimer's disease patients? A. Atropine B. Scopolamine (Hyoscine) C. Glycopyrrolate D. Ipratropium	Ans: C
Q8. The primary anesthetic concern in epilepsy is: A. Avoiding all inhalational agents B. Perioperative seizure prevention and drug interactions C. Mandatory use of regional anesthesia D. Avoiding opioids completely	Ans: B
Q9. Which anesthetic agent has the HIGHEST seizure-provoking potential? A. Propofol B. Sevoflurane C. Enflurane D. Isoflurane	Ans: C
Q10. Chronic phenytoin therapy affects neuromuscular blocking agents by: A. Prolonging the action of non-depolarizing agents B. Causing resistance to non-depolarizing agents C. Prolonging the action of succinylcholine D. Having no effect on muscle relaxants	Ans: B
Q11. Which intravenous anesthetic has anticonvulsant properties and is suitable for epileptic patients? A. Ketamine B. Etomidate C. Thiopentone D. Methohexitone	Ans: C
Q12. Status epilepticus is defined as continuous seizure activity lasting more than: A. 2 minutes B. 5 minutes C. 15 minutes D. 30 minutes	Ans: B
Q13. In a patient with acute stroke presenting for emergency surgery, the most important anesthetic goal is: A. Controlled hypotension B. Maintenance of cerebral perfusion pressure C. Deep anesthesia to reduce CMRO₂ D. Induced hypothermia to 32°C	Ans: B

Q14. Elective surgery should ideally be postponed for how long after an ischemic stroke? A. 2 weeks B. 4 weeks C. 6 weeks D. 3 months	Ans: C
Q15. Which of the following is TRUE regarding anesthesia for carotid endarterectomy in stroke patients? A. General anesthesia is always superior to regional B. Regional anesthesia allows neurological monitoring during cross-clamp C. Hypotension is desirable during cross-clamping D. Hypothermia below 33°C is mandatory	Ans: B
Q16. In patients with chronic headache/migraine, which of the following agents may trigger an attack? A. Propofol B. Nitroglycerin (GTN) C. Fentanyl D. Lidocaine	Ans: B
Q17. Post-dural puncture headache is characteristically: A. Worse when lying down B. Worse when sitting or standing (postural) C. Associated with fever D. Unilateral and throbbing	Ans: B
Q18. Multiple Sclerosis is characterized by: A. Destruction of anterior horn cells B. Demyelination in the central nervous system C. Peripheral nerve demyelination D. Neuromuscular junction dysfunction	Ans: B
Q19. In Multiple Sclerosis, which of the following may cause relapse/exacerbation? A. Hypothermia B. Hyperthermia (fever/increased body temperature) C. General anesthesia always D. Regional anesthesia always	Ans: B
Q20. Regarding regional anesthesia in Multiple Sclerosis: A. Epidural anesthesia is absolutely contraindicated B. Spinal anesthesia may worsen symptoms due to direct exposure of demyelinated cord to local anesthetic C. Peripheral nerve blocks are contraindicated D. All regional techniques are completely safe	Ans: B

Q21. Succinylcholine should be AVOIDED in Multiple Sclerosis because of: A. Malignant hyperthermia risk B. Hyperkalemia from denervation-type receptor changes C. Prolonged neuromuscular block D. Anaphylaxis risk	Ans: B
Q22. Syringomyelia is characterized by: A. Cavity formation in the cerebral cortex B. Fluid-filled cavity (syrinx) within the spinal cord C. Demyelination of peripheral nerves D. Inflammation of the meninges	Ans: B
Q23. The classic sensory finding in syringomyelia is: A. Glove and stocking sensory loss B. Dissociated sensory loss (loss of pain/temperature with preserved touch) C. Complete loss of all sensations below lesion D. Loss of proprioception only	Ans: B
Q24. Anesthetic concern in syringomyelia includes: A. Difficult airway due to limited neck extension B. Mandatory use of spinal anesthesia C. Succinylcholine is safe to use D. Temperature monitoring is not needed	Ans: A
Q25. Amyotrophic Lateral Sclerosis (ALS) affects: A. Only upper motor neurons B. Only lower motor neurons C. Both upper and lower motor neurons D. Only sensory neurons	Ans: C
Q26. In ALS patients, succinylcholine is contraindicated because of: A. Risk of malignant hyperthermia B. Life-threatening hyperkalemia C. Prolonged apnea D. Histamine release	Ans: B
Q27. The most important postoperative concern in ALS patients is: A. Pain management B. Respiratory failure requiring prolonged ventilation C. Urinary retention D. Hypothermia	Ans: B
Q28. ALS patients show increased sensitivity to: A. Inhalational agents only B. Non-depolarizing neuromuscular blocking agents C. Local anesthetics D. Antiemetics	Ans: B

Q29. Guillain-Barré Syndrome (GBS) is characterized by: A. Ascending central demyelination B. Acute inflammatory demyelinating polyradiculoneuropathy (peripheral) C. Anterior horn cell destruction D. Neuromuscular junction antibodies	Ans: B
Q30. The most dangerous complication of GBS requiring ICU admission is: A. Renal failure B. Respiratory failure from diaphragmatic weakness C. Seizures D. Hepatic failure	Ans: B
Q31. In GBS, succinylcholine is contraindicated due to: A. Prolonged block only B. Risk of hyperkalemia from denervation C. Histamine release D. Bradycardia	Ans: B
Q32. Autonomic dysfunction in GBS may manifest as: A. Only hypertension B. Labile blood pressure, tachycardia, bradycardia, cardiac arrest C. Only bradycardia D. Only urinary retention	Ans: B
Q33. Autonomic dysfunction (dysautonomia) is characterized by: A. Stable hemodynamics B. Failure of the autonomic nervous system to regulate BP, HR, and temperature C. Only parasympathetic overactivity D. Only sympathetic overactivity	Ans: B
Q34. In patients with autonomic dysfunction, the anesthetic concern includes: A. Exaggerated hypotensive response to anesthetic agents B. Need for higher doses of vasopressors C. Resistance to all anesthetic agents D. Increased risk of bronchospasm	Ans: A
Q35. Orthostatic hypotension in autonomic dysfunction is defined as a fall in systolic BP of: A. >10 mmHg on standing B. >20 mmHg on standing C. >30 mmHg on standing D. >40 mmHg on standing	Ans: B

Q36. Spinal cord transection above T6 may result in: A. Hypothermia only B. Autonomic hyperreflexia C. Hyperthermia only D. No cardiovascular changes	Ans: B
Q37. The most common trigger for autonomic hyperreflexia is: A. Emotional stress B. Bladder distension C. Cold environment D. Exercise	Ans: B
Q38. Treatment of autonomic hyperreflexia includes all EXCEPT: A. Remove the stimulus B. Sit the patient upright C. Administer vasopressors D. Give vasodilators (GTN/nifedipine)	Ans: C
Q39. In acute spinal cord injury (spinal shock phase), the hemodynamic picture is: A. Hypertension and tachycardia B. Hypotension and bradycardia (neurogenic shock) C. Normal blood pressure with tachycardia D. Hypertension and bradycardia	Ans: B
Q40. Succinylcholine is safe to use after spinal cord injury only within: A. First 24 hours B. First 48 hours C. First 72 hours D. First week	Ans: B
Q41. Which psychiatric medication has the most significant anesthetic interaction? A. SSRIs B. MAO inhibitors C. Benzodiazepines D. Atypical antipsychotics	Ans: B
Q42. Lithium therapy affects anesthesia by: A. Shortening the action of muscle relaxants B. Prolonging the action of both depolarizing and non-depolarizing muscle relaxants C. Having no effect on anesthetic drugs D. Reducing the MAC of volatile agents	Ans: B

Q43. Tricyclic antidepressants (TCAs) may cause which of the following during anesthesia? A. Exaggerated response to indirect sympathomimetics (ephedrine) B. Resistance to vasopressors C. Bronchospasm D. Malignant hyperthermia	Ans: A
Q44. Electroconvulsive therapy (ECT) requires anesthesia with: A. Deep general anesthesia with intubation B. Brief general anesthesia with short-acting agents (methohexitone/propofol + succinylcholine) C. Spinal anesthesia D. Local anesthesia only	Ans: B
Q45. Neuroleptic Malignant Syndrome (NMS) is characterized by all EXCEPT: A. Hyperthermia B. Muscle rigidity C. Hypotension and hypothermia D. Autonomic instability	Ans: C
Q46. In a patient with Parkinson's disease, which anesthetic technique may be preferred for lower limb surgery? A. General anesthesia with enflurane B. Regional anesthesia (spinal/epidural) C. Deep sedation with ketamine D. Only local infiltration	Ans: B
Q47. In Multiple Sclerosis, which factor does NOT increase the risk of perioperative relapse? A. Infection/fever B. Emotional stress C. Propofol administration D. Surgery itself	Ans: C
Q48. Guillain-Barré Syndrome typically presents how many weeks after a respiratory or GI infection? A. 1–4 weeks B. 6–8 weeks C. 3 months D. 6 months	Ans: A
Q49. In autonomic dysfunction, which monitoring is ESSENTIAL during anesthesia? A. Only pulse oximetry B. Invasive arterial blood pressure monitoring C. Only NIBP every 15 minutes D. BIS monitoring	Ans: B

Q50. Which anticonvulsant medication is associated with aplastic anemia, requiring preoperative blood count check?

- A. Phenytoin
- B. Carbamazepine
- C. Sodium valproate
- D. Gabapentin

Ans: B

Answer Key & Explanations

Q1. Answer: B — Levodopa/Carbidopa

Correct Answer: B. Levodopa/Carbidopa — Levodopa should be continued perioperatively to prevent rigidity, tremor, and difficulty with ventilation. Abrupt withdrawal can cause neuroleptic malignant-like syndrome.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q2. Answer: C — Droperidol

Correct Answer: C. Droperidol — Droperidol and other butyrophenones have antidopaminergic activity and can worsen parkinsonian symptoms. Metoclopramide should also be avoided for the same reason.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q3. Answer: B — Aspiration pneumonia

Correct Answer: B. Aspiration pneumonia — Parkinsonian patients have swallowing dysfunction and pharyngeal muscle rigidity leading to increased risk of aspiration. Rapid sequence induction may be considered.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q4. Answer: B — Meperidine (Pethidine) causing serotonin syndrome

Correct Answer: B. Meperidine (Pethidine) causing serotonin syndrome — MAO inhibitors with meperidine can cause fatal serotonin syndrome (hyperthermia, rigidity, seizures). Meperidine must be avoided in these patients.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q5. Answer: B — Anticholinesterase inhibitors (Donepezil)

Correct Answer: B. Anticholinesterase inhibitors (Donepezil) — Donepezil inhibits cholinesterase, which can prolong the duration of succinylcholine and antagonize non-depolarizing neuromuscular blocking agents.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

Q6. Answer: B — Postoperative cognitive dysfunction (POCD)

Correct Answer: B. Postoperative cognitive dysfunction (POCD) — Alzheimer's patients are at increased risk of POCD and postoperative delirium. Centrally acting anticholinergics (atropine) should be avoided.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

Q7. Answer: C — Glycopyrrolate

Correct Answer: C. Glycopyrrolate — Glycopyrrolate does not cross the blood-brain barrier and is preferred over atropine or scopolamine, which can worsen confusion and cognitive dysfunction in Alzheimer's patients.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

Q8. Answer: B — Perioperative seizure prevention and drug interactions

Correct Answer: B. Perioperative seizure prevention and drug interactions — Antiepileptic drugs must be continued perioperatively. Many anticonvulsants induce hepatic enzymes, altering metabolism of anesthetic drugs.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q9. Answer: C — Enflurane

Correct Answer: C. Enflurane — Enflurane at high concentrations (>2 MAC) especially with hypocapnia produces epileptiform EEG activity. It should be avoided in epileptic patients.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q10. Answer: B — Causing resistance to non-depolarizing agents

Correct Answer: B. Causing resistance to non-depolarizing agents — Chronic anticonvulsant therapy (phenytoin, carbamazepine) causes resistance to non-depolarizing muscle relaxants due to hepatic enzyme induction and upregulation of acetylcholine receptors.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q11. Answer: C — Thiopentone

Correct Answer: C. Thiopentone — Thiopentone and propofol have anticonvulsant properties. Methohexitone and ketamine may provoke seizures and should be avoided in epileptic patients.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q12. Answer: B — 5 minutes

Correct Answer: B. 5 minutes — Current definition: continuous seizure activity ≥ 5 minutes or ≥ 2 seizures without full recovery between them. First-line treatment is IV benzodiazepines (diazepam/lorazepam).

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q13. Answer: B — Maintenance of cerebral perfusion pressure

Correct Answer: B. Maintenance of cerebral perfusion pressure — In stroke patients, the ischemic penumbra depends on adequate perfusion. Avoiding hypotension and maintaining CPP is the primary anesthetic goal.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q14. Answer: C — 6 weeks

Correct Answer: C. 6 weeks — Elective surgery should be delayed at least 6 weeks after a stroke to allow stabilization of the blood-brain barrier and reduction of hemorrhagic transformation risk.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q15. Answer: B — Regional anesthesia allows neurological monitoring during cross-clamp

Correct Answer: B. Regional anesthesia allows neurological monitoring during cross-clamp — Cervical plexus block allows continuous neurological assessment during carotid cross-clamping to detect cerebral ischemia.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q16. Answer: B — Nitroglycerin (GTN)

Correct Answer: B. Nitroglycerin (GTN) — Nitroglycerin is a potent vasodilator and well-known migraine trigger. It should be used cautiously in patients with a history of migraines.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

Q17. Answer: B — Worse when sitting or standing (postural)

Correct Answer: B. Worse when sitting or standing (postural) — PDPH is a postural headache that worsens in the upright position and improves on lying flat, caused by CSF leak and reduced intracranial pressure.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

Q18. Answer: B — Demyelination in the central nervous system

Correct Answer: B. Demyelination in the central nervous system — MS is an autoimmune disease causing patchy demyelination in the brain and spinal cord, leading to relapsing-remitting neurological deficits.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q19. Answer: B — Hyperthermia (fever/increased body temperature)

Correct Answer: B. Hyperthermia (fever/increased body temperature) — Even small increases in body temperature can worsen symptoms in MS (Uhthoff's phenomenon). Perioperative temperature control is essential.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q20. Answer: B — Spinal anesthesia may worsen symptoms due to direct exposure of demyelinated cord to local anesthetic

Correct Answer: B. Spinal anesthesia may worsen symptoms due to direct exposure of demyelinated cord to local anesthetic — Spinal anesthesia exposes the already demyelinated cord to local anesthetic. Epidural is relatively safer; lowest effective concentration should be used.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q21. Answer: B — Hyperkalemia from denervation-type receptor changes

Correct Answer: B. Hyperkalemia from denervation-type receptor changes — Upper motor neuron lesions in MS lead to extrajunctional acetylcholine receptor proliferation, causing hyperkalemic response to succinylcholine.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q22. Answer: B — Fluid-filled cavity (syrinx) within the spinal cord

Correct Answer: B. Fluid-filled cavity (syrinx) within the spinal cord — Syringomyelia is a condition where a fluid-filled cyst (syrinx) forms within the spinal cord, often in the cervical region.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

Q23. Answer: B — Dissociated sensory loss (loss of pain/temperature with preserved touch)

Correct Answer: B. Dissociated sensory loss (loss of pain/temperature with preserved touch) — The syrinx destroys the crossing spinothalamic fibers causing cape-like loss of pain and temperature while posterior column sensation (touch, proprioception) is preserved.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

Q24. Answer: A — Difficult airway due to limited neck extension

Correct Answer: A. Difficult airway due to limited neck extension — Syringomyelia often coexists with Arnold-Chiari malformation causing limited neck mobility. Careful airway management with minimal neck manipulation is essential.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

Q25. Answer: C — Both upper and lower motor neurons

Correct Answer: C. Both upper and lower motor neurons — ALS (Motor Neuron Disease) is a progressive neurodegenerative disease affecting both upper and lower motor neurons, leading to muscle weakness, wasting, and spasticity.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

Q26. Answer: B — Life-threatening hyperkalemia

Correct Answer: B. Life-threatening hyperkalemia — Lower motor neuron involvement in ALS causes denervation and extrajunctional receptor proliferation, making succinylcholine extremely dangerous due to massive potassium release.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

Q27. Answer: B — Respiratory failure requiring prolonged ventilation

Correct Answer: B. Respiratory failure requiring prolonged ventilation — ALS patients have progressive respiratory muscle weakness. They are extremely sensitive to respiratory depressants and may require postoperative ventilatory support.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

Q28. Answer: B — Non-depolarizing neuromuscular blocking agents

Correct Answer: B. Non-depolarizing neuromuscular blocking agents — Due to decreased muscle mass and denervation, ALS patients are very sensitive to non-depolarizing agents. Reduced doses should be used with careful monitoring.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

Q29. Answer: B — Acute inflammatory demyelinating polyradiculoneuropathy (peripheral)

Correct Answer: B. Acute inflammatory demyelinating polyradiculoneuropathy (peripheral) — GBS is an autoimmune peripheral nerve demyelination causing ascending flaccid paralysis, often post-infectious.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

Q30. Answer: B — Respiratory failure from diaphragmatic weakness

Correct Answer: B. Respiratory failure from diaphragmatic weakness — Ascending paralysis may involve respiratory muscles. FVC <20 mL/kg, NIF <-25 cmH₂O, or >30% decline in FVC indicates need for intubation.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

Q31. Answer: B — Risk of hyperkalemia from denervation

Correct Answer: B. Risk of hyperkalemia from denervation — Succinylcholine should be avoided from the onset of GBS due to risk of hyperkalemia from denervation-induced upregulation of acetylcholine receptors at the neuromuscular junction.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

Q32. Answer: B — Labile blood pressure, tachycardia, bradycardia, cardiac arrest

Correct Answer: B. Labile blood pressure, tachycardia, bradycardia, cardiac arrest — Autonomic instability in GBS can cause unpredictable hemodynamic swings. Invasive monitoring is recommended.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

Q33. Answer: B — Failure of the autonomic nervous system to regulate BP, HR, and temperature

Correct Answer: B. Failure of the autonomic nervous system to regulate BP, HR, and temperature — Autonomic dysfunction involves both sympathetic and parasympathetic failure, causing orthostatic hypotension, temperature dysregulation, and labile hemodynamics.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

Q34. Answer: A — Exaggerated hypotensive response to anesthetic agents

Correct Answer: A. Exaggerated hypotensive response to anesthetic agents — Loss of compensatory sympathetic reflexes causes profound hypotension with induction agents, neuraxial blocks, and positive pressure ventilation.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

Q35. Answer: B — >20 mmHg on standing

Correct Answer: B. >20 mmHg on standing — Orthostatic hypotension is a sustained reduction of ≥20 mmHg systolic or ≥10 mmHg diastolic within 3 minutes of standing, indicating autonomic failure.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

Q36. Answer: B — Autonomic hyperreflexia

Correct Answer: B. Autonomic hyperreflexia — Lesions above T6 can trigger autonomic hyperreflexia: massive sympathetic discharge below the lesion causing severe hypertension, reflex bradycardia, headache, and flushing above the lesion.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q37. Answer: B — Bladder distension

Correct Answer: B. Bladder distension — Bladder distension (full catheter, bladder surgery) is the most common trigger. Other triggers include bowel distension, skin stimulation below the lesion level.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q38. Answer: C — Administer vasopressors

Correct Answer: C. Administer vasopressors — Vasopressors would worsen the hypertension. Treatment: remove stimulus, sit up patient, give vasodilators (GTN sublingual/nifedipine), deepen anesthesia if intraoperative.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q39. Answer: B — Hypotension and bradycardia (neurogenic shock)

Correct Answer: B. Hypotension and bradycardia (neurogenic shock) — Acute spinal cord injury causes loss of sympathetic tone below the lesion resulting in vasodilation (hypotension) and unopposed vagal tone (bradycardia).

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q40. Answer: B — First 48 hours

Correct Answer: B. First 48 hours — Succinylcholine is relatively safe within the first 48 hours after spinal cord injury. After that, denervation causes receptor upregulation and life-threatening hyperkalemia risk, lasting for 6-9 months or more.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q41. Answer: B — MAO inhibitors

Correct Answer: B. MAO inhibitors — MAOIs have the most dangerous interactions: hypertensive crisis with indirect sympathomimetics, serotonin syndrome with meperidine/tramadol. Traditionally stopped 2 weeks preoperatively.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q42. Answer: B — Prolonging the action of both depolarizing and non-depolarizing muscle relaxants

Correct Answer: B. Prolonging the action of both depolarizing and non-depolarizing muscle relaxants — Lithium inhibits acetylcholine synthesis and release, potentiating all neuromuscular blocking agents. It also reduces renal concentrating ability (diabetes insipidus-like).

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q43. Answer: A — Exaggerated response to indirect sympathomimetics (ephedrine)

Correct Answer: A. Exaggerated response to indirect sympathomimetics (ephedrine) — TCAs block norepinephrine reuptake, potentiating the action of both direct and indirect sympathomimetics. Arrhythmias and hypertensive crisis may occur.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q44. Answer: B — Brief general anesthesia with short-acting agents (methohexitone/propofol + succinylcholine)

Correct Answer: B. Brief general anesthesia with short-acting agents (methohexitone/propofol + succinylcholine) — ECT requires brief GA to prevent injuries from seizure-induced muscle contraction. Methohexitone is traditional; propofol may reduce seizure duration.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q45. Answer: C — Hypotension and hypothermia

Correct Answer: C. Hypotension and hypothermia — NMS presents with hyperthermia, severe rigidity, altered consciousness, autonomic instability (tachycardia, labile BP), and elevated CK. Treatment includes dantrolene and dopamine agonists.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q46. Answer: B — Regional anesthesia (spinal/epidural)

Correct Answer: B. Regional anesthesia (spinal/epidural) — Regional anesthesia avoids drug interactions, aspiration risk, and respiratory depression. It is often preferred when feasible in Parkinsonian patients.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q47. Answer: C — Propofol administration

Correct Answer: C. Propofol administration — There is no evidence that propofol triggers MS relapse. However, infection, fever, stress, and the physiological stress of surgery are known triggers for MS exacerbation.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q48. Answer: A — 1–4 weeks

Correct Answer: A. 1–4 weeks — GBS typically develops 1–4 weeks after a viral illness (commonly *Campylobacter jejuni*, CMV, EBV). It presents as ascending symmetrical weakness with areflexia.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

Q49. Answer: B — Invasive arterial blood pressure monitoring

Correct Answer: B. Invasive arterial blood pressure monitoring — Due to labile hemodynamics and exaggerated responses to anesthetic drugs, continuous invasive BP monitoring is essential for titration of drugs and fluids.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

Q50. Answer: B — Carbamazepine

Correct Answer: B. Carbamazepine — Carbamazepine can cause aplastic anemia and agranulocytosis. Preoperative full blood count is essential. Sodium valproate can cause thrombocytopenia and platelet dysfunction.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy